

gdTAI V4 Final Evaluation

Development-frozen training, common external lockbox comparison, and final no-promotion decision

Decision: do not promote gdTAI V4.3. V4.3 met its false-positive limits but failed external recall and did not outperform V3. The current V3 balanced model remains the default. No V4 artifact was released, pushed, or applied to the 5.9-million-cell atlas.

45.62% V4.3 external gdT recall	0.24% V4.3 paired-alpha-beta FPR	0.76% V4.3 author-NK FPR	0.619 V4.3 external F1
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1. Evaluation Design

The final V4.3 candidate, all thresholds, receptor-rescue rules, and the feature contract were frozen using development cells only. The external lockbox was checksum-bound before scoring and contained **335,479 cells**: 41,931 gdT gold, 285,524 paired-alpha-beta gold, and 8,024 independent author-NK negatives. V4.3, V3, and both V2 modes were recomputed from the same raw-count matrix and identical cells.

Ground truth. gdT positives were sorted gdT cohorts or strict paired productive TRG/TRD cells without alpha-beta receptor evidence. Alpha-beta negatives had productive alpha/beta receptor evidence without gamma/delta evidence. Author-NK negatives had NK annotation and no productive TCR. Where chain UMI support existed, one-UMI calls were not accepted as gold; datasets without UMI fields retained published productive calls.

V4.3 architecture: a 36-gene high-recall T-lineage support gate followed by a 158-gene receptor classifier. Stage 2 uses 153 individual TCR genes plus `CD3D/E/G` , `CD4` , and `FOXP3` . `KLRD1` , `NKG7` , `GNLY` , `FCER1G` , `TYROBP` , and the shared cytotoxic program are absent from both stages. Stage 1 probability is not supplied to Stage 2.

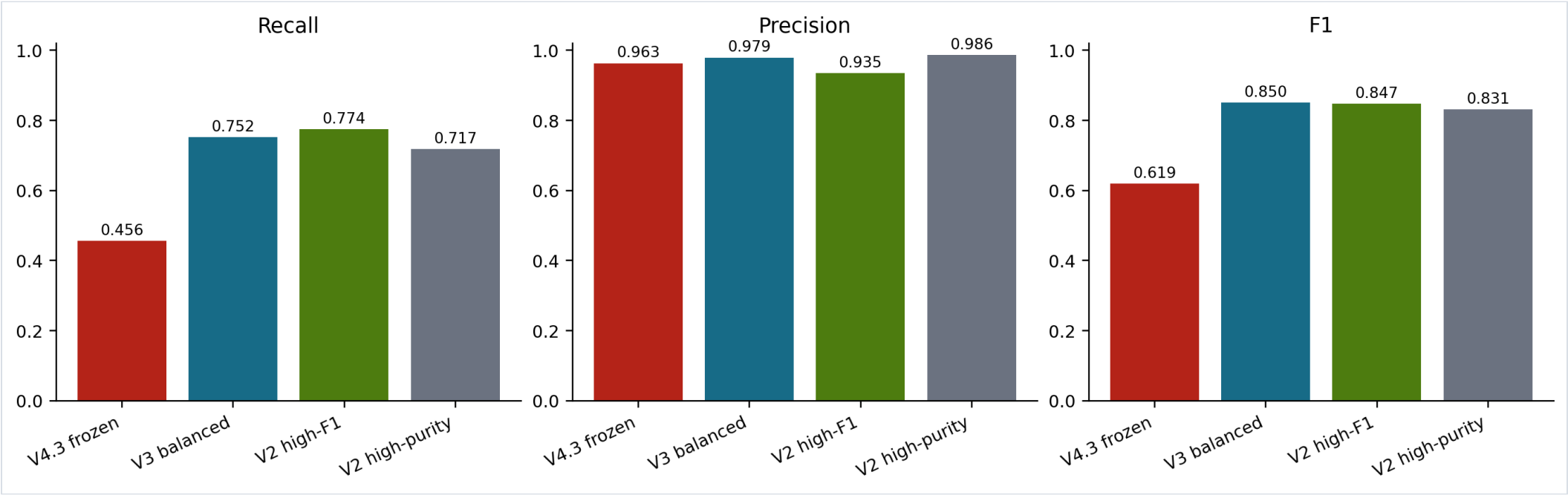
Implementation integrity

check	value	result
lockbox cells matching original frozen cache	327455.000	PASS
new author-NK cells requiring direct extraction	8024.000	EXPECTED
effective V4 genes compared	183.000	PASS

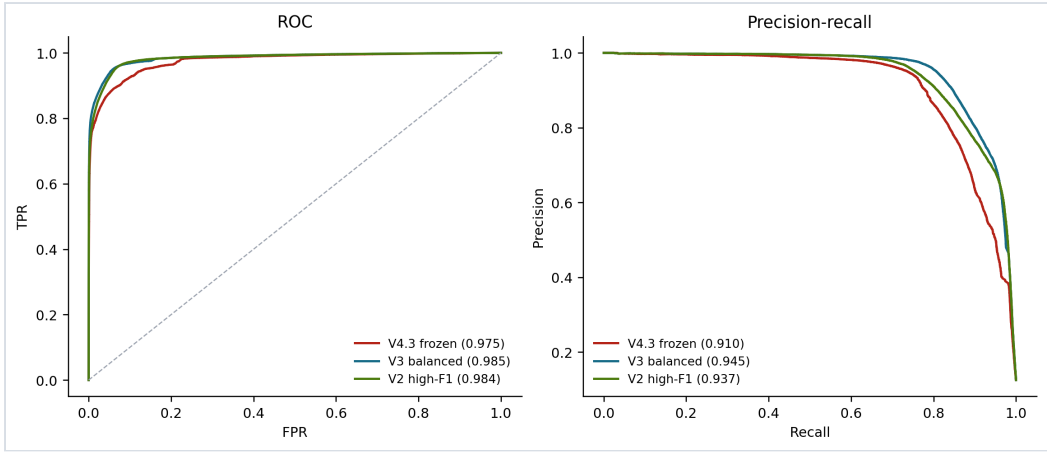
check	value	result
feature values compared	59924265.000	PASS
values differing by >1e-6	0.000	PASS
maximum absolute feature difference	0.000	PASS

2. Headline Performance

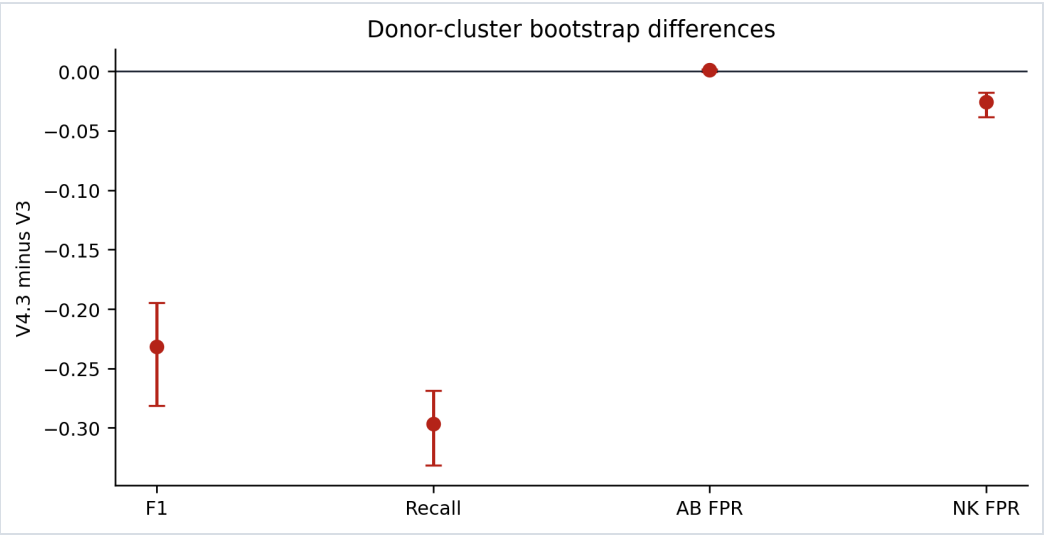
model	predicted_positive	precision	recall	f1	abt_fpr	author_nk_fpr	precision_at_1pct_prevalence
V4.3 frozen	19872	96.25%	45.62%	61.90%	0.24%	0.76%	64.48%
V3 balanced	32210	97.87%	75.18%	85.04%	0.15%	3.35%	76.47%
V2 high-F1	34744	93.45%	77.44%	84.70%	0.20%	21.07%	50.24%
V2 high-purity	30501	98.63%	71.74%	83.06%	0.10%	1.68%	83.54%



V4.3 gained NK specificity relative to V3 but lost too much gdT recall. Precision at 1% prevalence is included because the atlas is strongly imbalanced.



V4.3 has lower ROC-AUC and average precision than V3 and V2; this is a ranking deficit, not only threshold calibration.



Donor-cluster bootstrap. The V4.3-minus-V3 F1 and recall intervals remain entirely below zero.

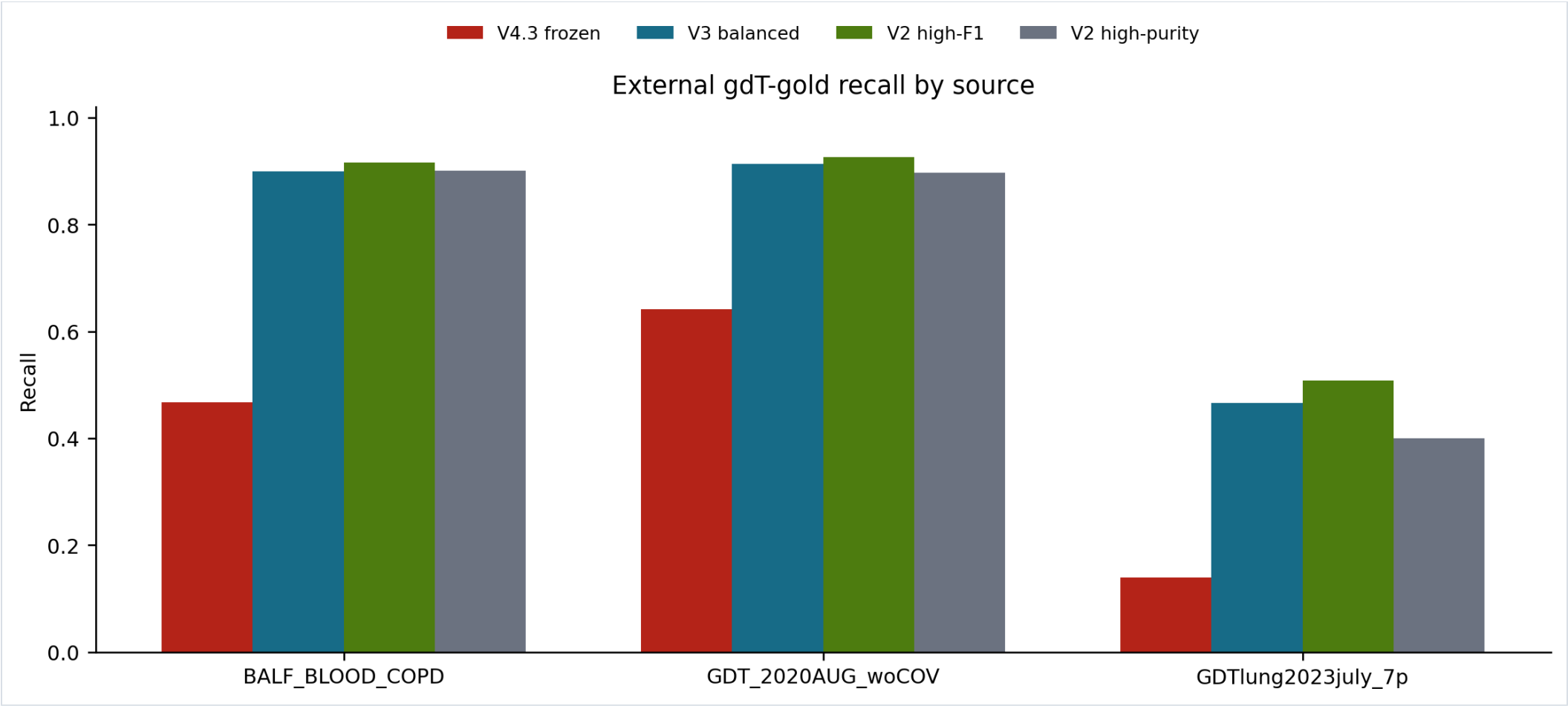
Ranking metrics

model	roc_auc	average_precision
V4.3 frozen	0.975	0.910
V3 balanced	0.985	0.945
V2 high-F1	0.984	0.937

Donor-cluster bootstrap

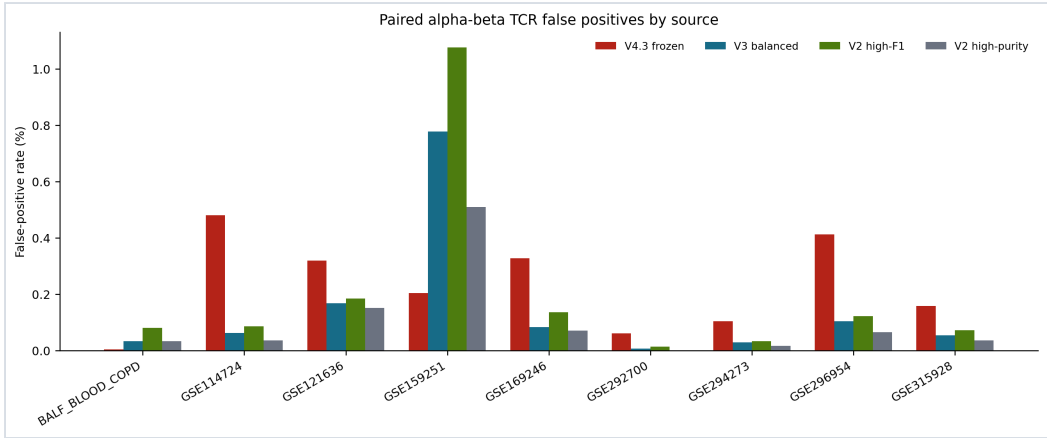
older_model	delta_f1_median	delta_f1_ci_low	delta_f1_ci_high	probability_delta_f1_gt_0	delta_recall_median	delta_recall_ci_low	delta_recall_ci_high
V3 balanced	-0.232	-0.281	-0.195	0.000	-0.296	-0.332	-0.269
V2 high-F1	-0.228	-0.283	-0.187	0.000	-0.319	-0.356	-0.288
V2 high-purity	-0.212	-0.254	-0.180	0.000	-0.261	-0.297	-0.239

3. Dataset-by-Dataset Results

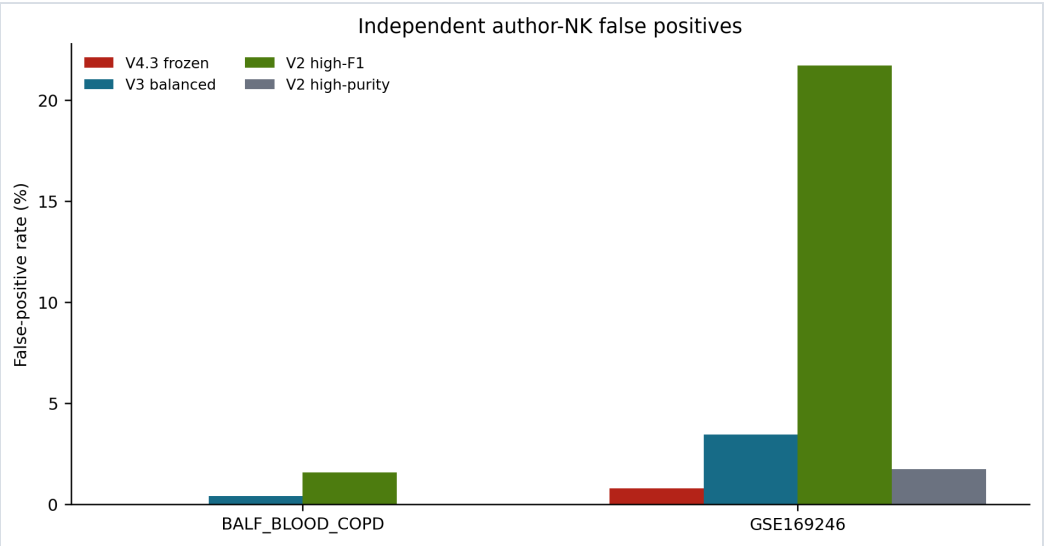


V4.3 under-recovers every locked positive source. GDTlung is retained as a documented low-quality stress cohort, but BALF_BLOOD_COPD and GDT_2020 also show large deficits.

source_gse_id	V4.3 frozen	V3 balanced	V2 high-F1	V2 high-purity
BALF_BLOOD_COPD	46.71%	89.91%	91.55%	90.02%
GDT_2020AUG_woCOV	64.15%	91.40%	92.57%	89.74%
GDTlung2023july_7p	13.92%	46.66%	50.81%	40.00%



Paired-alpha-beta false-positive rates vary by external source. V4.3 is not uniformly more specific than V3.



V4.3 reduces author-NK false positives relative to V3 and V2, but purity does not compensate for the recall collapse.

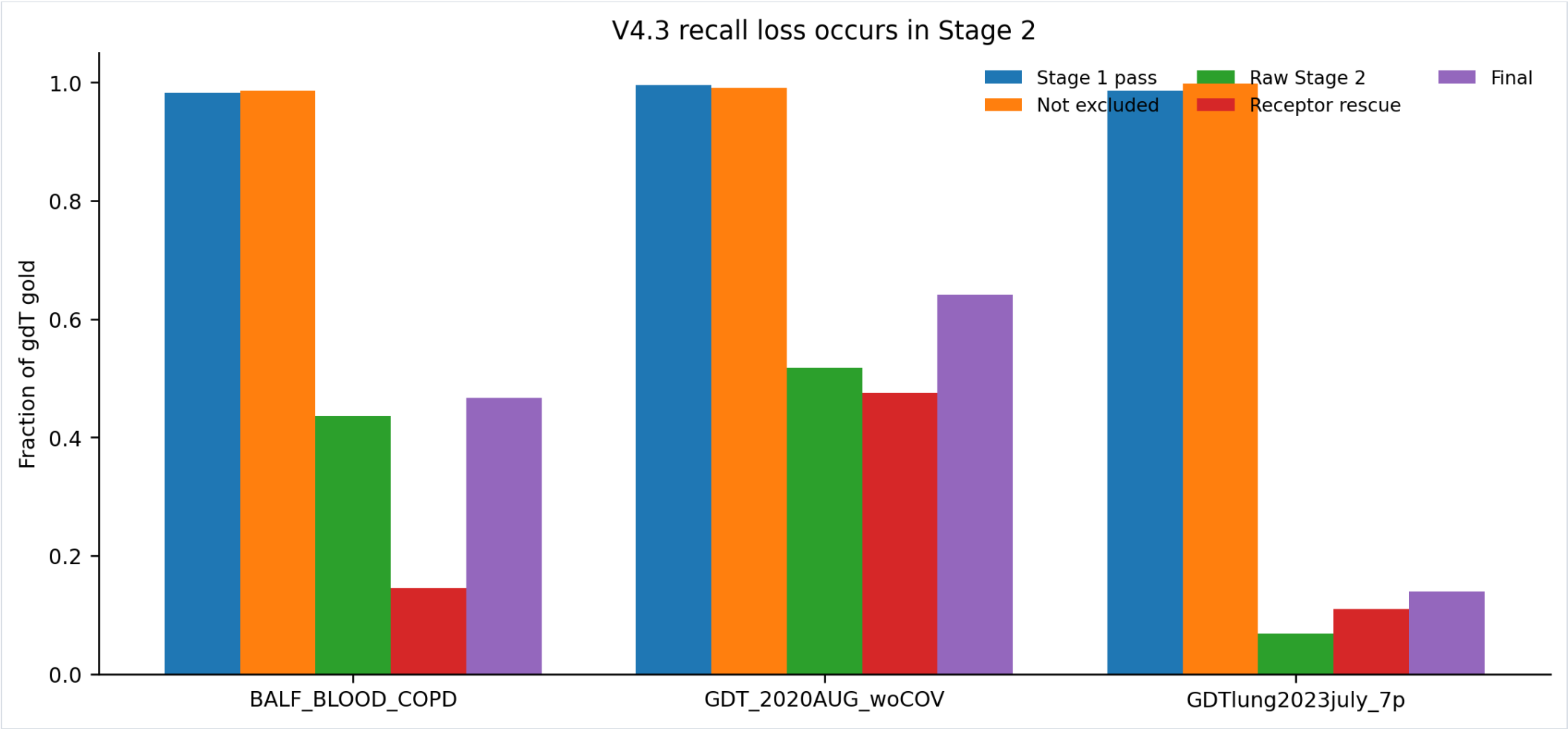
Paired-alpha-beta false-positive rates

source_gse_id	V4.3 frozen	V3 balanced	V2 high-F1	V2 high-purity
BALF_BLOOD_COPD	0.00%	0.03%	0.08%	0.03%
GSE114724	0.48%	0.06%	0.09%	0.04%
GSE121636	0.32%	0.17%	0.18%	0.15%
GSE159251	0.20%	0.78%	1.08%	0.51%
GSE169246	0.33%	0.08%	0.14%	0.07%
GSE292700	0.06%	0.01%	0.01%	0.00%
GSE294273	0.10%	0.03%	0.03%	0.02%
GSE296954	0.41%	0.10%	0.12%	0.07%
GSE315928	0.16%	0.05%	0.07%	0.04%

Author-NK false-positive rates

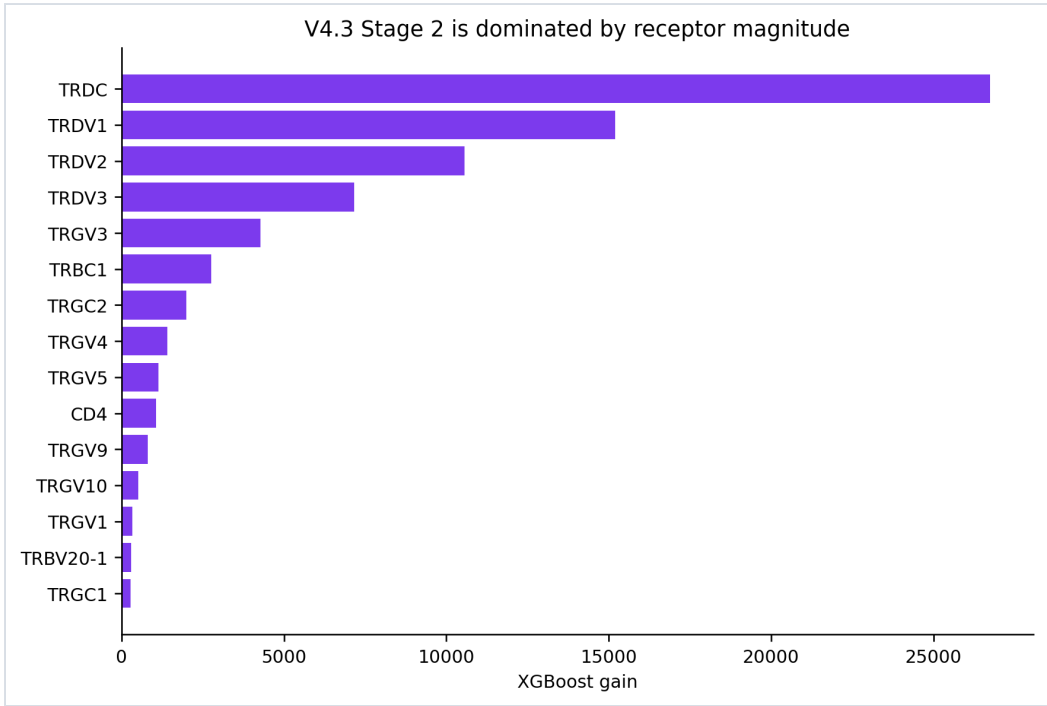
source_gse_id	V4.3 frozen	V3 balanced	V2 high-F1	V2 high-purity
BALF_BLOOD_COPD	0.00%	0.39%	1.57%	0.00%
GSE169246	0.79%	3.45%	21.71%	1.74%

4. Why V4.3 Failed

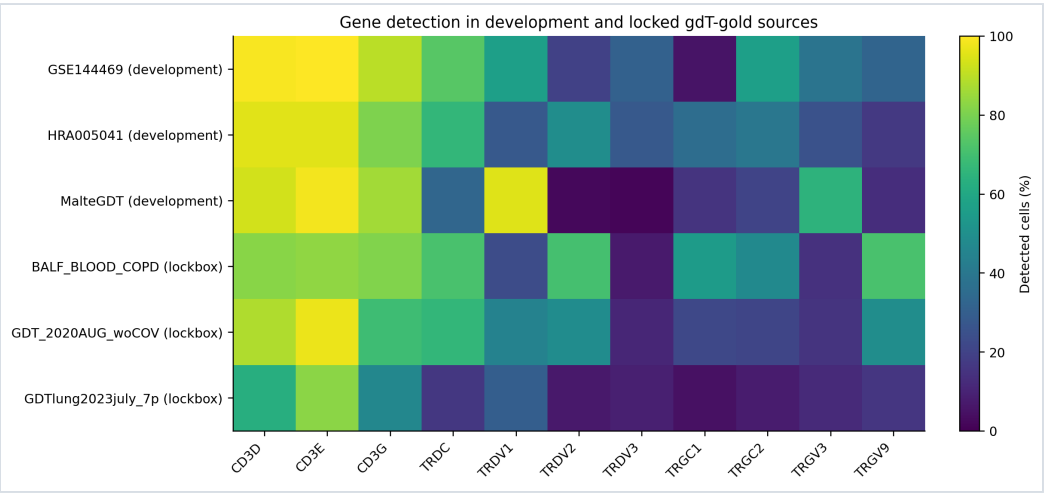


Stage 1 passes 98-100% of locked gdT gold and CD4/Treg exclusions affect no more than 1.4%. Most false negatives arise from the Stage 2 receptor classifier.

source_gse_id	n_gdT_gold	stage1_pass	not_cd4_treg_excluded	raw_stage2_pass	receptor_rescue	final_recall	v3_recall	v3_recovered_v4_missed
BALF_BLOOD_COPD	852	98.24%	98.59%	43.54%	14.55%	46.71%	89.91%	375
GDT_2020AUG_woCOV	25904	99.53%	99.07%	51.78%	47.56%	64.15%	91.40%	7332
GDTlung2023july_7p	15175	98.64%	99.76%	6.85%	10.95%	13.92%	46.66%	5023



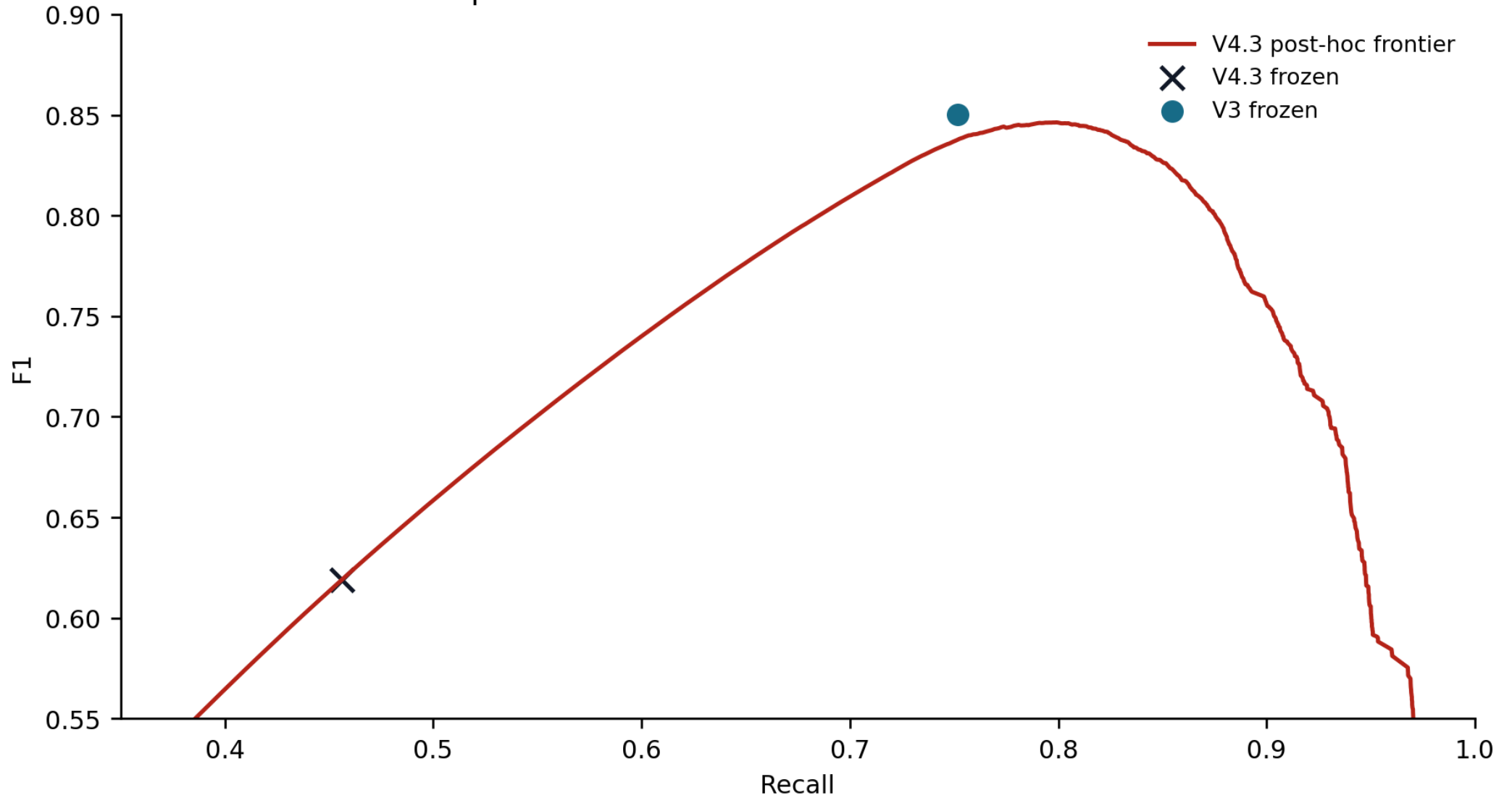
TRDC and TRDV magnitude dominate the nonlinear Stage 2 model despite the broader individual-receptor panel.



Positive sources differ substantially in receptor detection. GDTlung has especially weak TRDC/CD3 support, while BALF and GDT_2020 also differ from development cohorts.

The final Stage 2 threshold was `0.994098` , selected from grouped development OOF predictions. Lowering it after seeing the lockbox would be leakage. A diagnostic-only threshold sweep nevertheless shows whether calibration alone could solve the problem.

Even the post-hoc V4.3 threshold frontier does not exceed V3



The best post-hoc V4.3 F1 is still below frozen V3. Matching V3 recall requires substantially more alpha-beta false positives, so threshold adjustment cannot establish superiority.

diagnostic	threshold	f1	recall	precision	abt_fpr	author_nk_fpr
posthoc_maximum_f1	0.270	84.64%	79.60%	90.36%	0.94%	10.95%
minimum_70pct_recall	0.801	80.94%	70.00%	95.94%	0.38%	1.83%
match_v3_recall	0.562	83.77%	75.18%	94.58%	0.54%	3.38%

diagnostic	threshold	f1	recall	precision	abt_fpr	author_nk_fpr
minimum_80pct_recall	0.246	84.62%	80.00%	89.81%	1.02%	11.28%
frozen_operating_point	0.994	61.90%	45.62%	96.25%	0.24%	0.76%

5. Iteration Record

iteration	change	result
V4.1 GPU	Two-stage T/NK architecture	Stopped at Gate C; Stage 2 never ran
V4.2	Repaired TCR joins, expression-independent truth, seven-source NK negatives	Failed development transfer; Malte recall 36.5%
V4.3 initial	Receptor-isolated Stage 2; removed Stage 1 probability	Passed development, but KLRD1 remained in Stage 1
V4.3 final	Removed KLRD1/shared cytotoxic genes from both stages	Passed development; failed locked external superiority

V4 development addressed the major known risks in sequence: weak NK labels, unsafe TCR joins, source leakage, shared cytotoxic genes, continuous Stage 1 leakage, and source-heldout evaluation. V4.3 is the first iteration to reach a clean common lockbox comparison, and it fails there.

6. Final Scientific Conclusion

With the current datasets, no defensible V4 improvement over V3 was demonstrated. Training another candidate after inspecting this lockbox would convert the external test into a tuning set. The ranking deficit also shows that a threshold-only revision is insufficient. Therefore V4 development should stop here rather than optimize against the consumed validation cohorts.

This conclusion is scoped to the available evidence, not a claim that gdT classification can never improve. A future V4 attempt requires genuinely new independent positive and NK/alpha-beta cohorts, or a prespecified domain-generalization design with a new untouched final test. Until then:

- retain V3 balanced as the current default;
- retain V2 high-purity as a conservative annotation-assisted fallback;
- keep all V4.3 artifacts diagnostic and unpromoted;
- do not use lockbox-derived thresholds or feature changes in a release model.

Core hashes: V4 contract `05cb3c9cd9a94577849610263141b4ce499b564acc2dff77f4a3b2d274a09abe` ; lockbox `f997e24b7be6e98692ba770d163916cbb359009ba5a7c3dfa1d7db6b3824f151` .
Model promotion status: false.